

Factor Model Diagnostics

Systematic Analysis of Correlation Structure

Key Finding

Strong raw-return associations typically diminish significantly once systematic factor exposure is removed.

This reveals that factor loading is the dominant driver of cross-asset correlation.

Benchmark Universe

18 assets used to construct the factor model

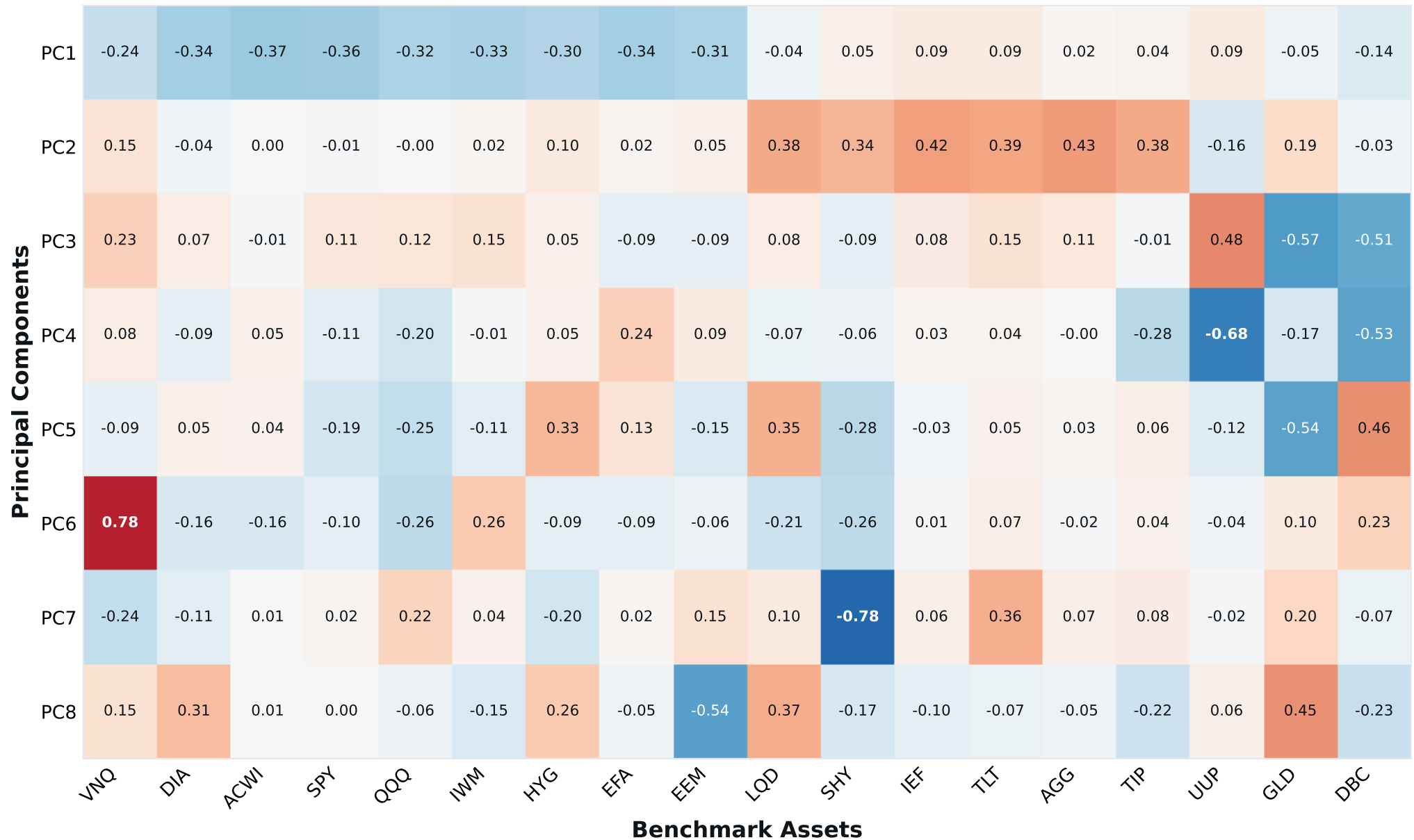
Training Period: January 04, 2012 to August 14, 2026 (3,674 observations)

ACWI Global equities	EEM Emerging-market equities	LQD Investment-grade credit
SPY US large-cap equities	AGG US aggregate bonds	HYG High-yield credit
DIA US industrial large-cap equities	SHY US 1-3 year Treasury bonds	VNQ US real estate
QQQ US large-cap growth equities	IEF US 7-10 year Treasury bonds	GLD Gold
IWM US small-cap equities	TLT US 20+ year Treasury bonds	DBC Broad commodities
EFA Developed ex-US equities	TIP US inflation-linked bonds	UUP US dollar

These benchmark assets form the fixed universe on which the factor model is trained. All factors are extracted from the daily returns of these instruments.

Factor Model Loadings

Principal component loadings on benchmark assets (first 8 PCs) — ordered by full PC similarity



Assets are ordered by hierarchical clustering of all PC loadings weighted by eigenvalues.
Dominant PCs (larger eigenvalues) guide similarity, capturing full factor structure.

Factor Model Methodology

Part 1: Model Construction

Objective

Decompose asset returns into systematic factor exposures and idiosyncratic residuals. The model captures co-movement driven by shared macroeconomic and market factors.

Construction Process

1. Input data: Daily returns of the benchmark universe and explanatory universe assets
2. Factor selection: Identify principal economic and market factors via PCA and correlation analysis
3. Factor fit: Non-linear regression of asset returns on selected factors (XGBoost ensemble)
4. Out-of-sample testing: Holdout period validation to assess generalization
5. Residual extraction: $\text{OOS residuals} = \text{actual returns} - \text{fitted factor predictions}$

Key Properties

- Factors are time-invariant and asset-universe-specific
- Model trained on benchmark + explanatory universe
- Residuals represent unexplained co-movement and asset-specific risk
- Out-of-sample validation prevents overfitting bias

Factor Model Methodology

Part 2: Diagnostic Application

Separate Analysis Universes

The factor model is constructed on one fixed universe. For diagnostic purposes, we apply this same model to a separate target universe of assets, measuring how well the fitted factors explain their behavior.

Analysis Workflow

1. Extract fitted factor loadings from model (trained universe)
2. Apply loadings to target universe assets
3. Compute target universe asset correlations (raw, daily returns)
4. Compute residual universe correlations (OOS fit residuals)
5. Compare correlation structures to identify factor-driven vs. unexplained co-movement

Diagnostic Insights

- If raw correlation $\gg$ residual correlation: co-movement driven by shared factors (model success)
- If raw correlation $\approx$ residual correlation: co-movement not captured by model (potential missing factors)
- Magnitude of change reveals the strength of factor-loading dependence

Target Universe

15 investible assets for diagnostic analysis

Analysis Period: March 23, 2022 to August 14, 2026 (1,103 observations)

PRPFX

MNA

VLUE

DBMF

CTA

JEPI

KMLM

MTUM

DIVO

BTAL

QUAL

ARKK

RPAR

USMV

QAI

These 15 assets comprise the target universe to which the factor model is applied. Pairwise analysis produces 105 unique correlations for diagnostic comparison.

Raw Returns

Pairwise Pearson Correlation of Daily Returns

PRPFX	1.00	0.13	-0.13	-0.54	0.68	0.40	-0.08	0.63	0.69	0.57	0.66	0.60	0.65	0.59	0.75
DBMF	0.13	1.00	0.57	-0.07	-0.18	0.01	0.41	0.20	0.04	-0.03	0.10	0.05	0.07	0.01	0.08
KMLM	-0.13	0.57	1.00	0.17	-0.34	-0.11	0.41	-0.06	-0.19	-0.19	-0.14	-0.13	-0.11	-0.23	-0.20
BTAL	-0.54	-0.07	0.17	1.00	-0.45	-0.30	0.14	-0.63	-0.62	-0.27	-0.63	-0.40	-0.42	-0.77	-0.68
RPAR	0.68	-0.18	-0.34	-0.45	1.00	0.37	-0.27	0.49	0.58	0.54	0.57	0.55	0.56	0.48	0.67
MNA	0.40	0.01	-0.11	-0.30	0.37	1.00	-0.07	0.36	0.45	0.45	0.46	0.45	0.44	0.39	0.46
CTA	-0.08	0.41	0.41	0.14	-0.27	-0.07	1.00	-0.09	-0.14	-0.14	-0.12	-0.12	-0.14	-0.13	-0.14
MTUM	0.63	0.20	-0.06	-0.63	0.49	0.36	-0.09	1.00	0.80	0.59	0.78	0.69	0.70	0.63	0.77
QUAL	0.69	0.04	-0.19	-0.62	0.58	0.45	-0.14	0.80	1.00	0.83	0.83	0.86	0.85	0.75	0.82
USMV	0.57	-0.03	-0.19	-0.27	0.54	0.45	-0.14	0.59	0.83	1.00	0.73	0.90	0.87	0.54	0.66
VLUE	0.66	0.10	-0.14	-0.63	0.57	0.46	-0.12	0.78	0.83	0.73	1.00	0.78	0.80	0.66	0.81
JEPI	0.60	0.05	-0.13	-0.40	0.55	0.45	-0.12	0.69	0.86	0.90	0.78	1.00	0.90	0.57	0.73
DIVO	0.65	0.07	-0.11	-0.42	0.56	0.44	-0.14	0.70	0.85	0.87	0.80	0.90	1.00	0.57	0.72
ARKK	0.59	0.01	-0.23	-0.77	0.48	0.39	-0.13	0.63	0.75	0.54	0.66	0.57	0.57	1.00	0.72
QAI	0.75	0.08	-0.20	-0.68	0.67	0.46	-0.14	0.77	0.82	0.66	0.81	0.73	0.72	0.72	1.00
PRPFX	DBMF	KMLM	BTAL	RPAR	MNA	CTA	MTUM	QUAL	USMV	VLUE	JEPI	DIVO	ARKK	QAI	

This matrix shows the baseline correlation structure before factor removal.

After Factor Removal

Pairwise Pearson Correlation of Out-of-Sample Residuals

PRPFX	1.00	-0.14	-0.14	-0.36	0.09	0.04	0.05	0.25	0.36	0.05	0.28	0.01	0.11	0.31	0.43
DBMF	-0.14	1.00	0.37	0.21	-0.17	-0.08	0.20	0.08	-0.23	-0.11	-0.18	0.07	-0.09	-0.21	-0.16
KMLM	-0.14	0.37	1.00	0.07	-0.02	0.02	0.09	-0.04	-0.09	-0.01	-0.01	0.06	0.01	-0.15	-0.19
BTAL	-0.36	0.21	0.07	1.00	-0.05	-0.01	0.02	-0.18	-0.12	0.43	-0.30	0.34	0.30	-0.59	-0.40
RPAR	0.09	-0.17	-0.02	-0.05	1.00	0.10	0.01	-0.03	0.16	0.15	0.28	0.08	0.27	0.02	0.03
MNA	0.04	-0.08	0.02	-0.01	0.10	1.00	0.05	-0.03	0.10	0.16	0.13	0.06	0.11	0.09	0.09
CTA	0.05	0.20	0.09	0.02	0.01	0.05	1.00	0.05	0.13	0.06	0.01	0.01	0.07	0.08	0.02
MTUM	0.25	0.08	-0.04	-0.18	-0.03	-0.03	0.05	1.00	0.23	0.02	0.28	0.13	0.06	0.03	0.27
QUAL	0.36	-0.23	-0.09	-0.12	0.16	0.10	0.13	0.23	1.00	0.54	0.30	0.36	0.40	0.26	0.27
USMV	0.05	-0.11	-0.01	0.43	0.15	0.16	0.06	0.02	0.54	1.00	0.21	0.62	0.62	-0.08	0.02
VLUE	0.28	-0.18	-0.01	-0.30	0.28	0.13	0.01	0.28	0.30	0.21	1.00	0.09	0.28	0.15	0.38
JEPI	0.01	0.07	0.06	0.34	0.08	0.06	0.01	0.13	0.36	0.62	0.09	1.00	0.46	-0.16	0.07
DIVO	0.11	-0.09	0.01	0.30	0.27	0.11	0.07	0.06	0.40	0.62	0.28	0.46	1.00	-0.15	-0.01
ARKK	0.31	-0.21	-0.15	-0.59	0.02	0.09	0.08	0.03	0.26	-0.08	0.15	-0.16	-0.15	1.00	0.35
QAI	0.43	-0.16	-0.19	-0.40	0.03	0.09	0.02	0.27	0.27	0.02	0.38	0.07	-0.01	0.35	1.00
PRPFX	DBMF	KMLM	BTAL	RPAR	MNA	CTA	MTUM	QUAL	USMV	VLUE	JEPI	DIVO	ARKK	QAI	

Notice the substantial reduction in correlation magnitude—evidence of factor model explanatory power.

Pairwise Transformations

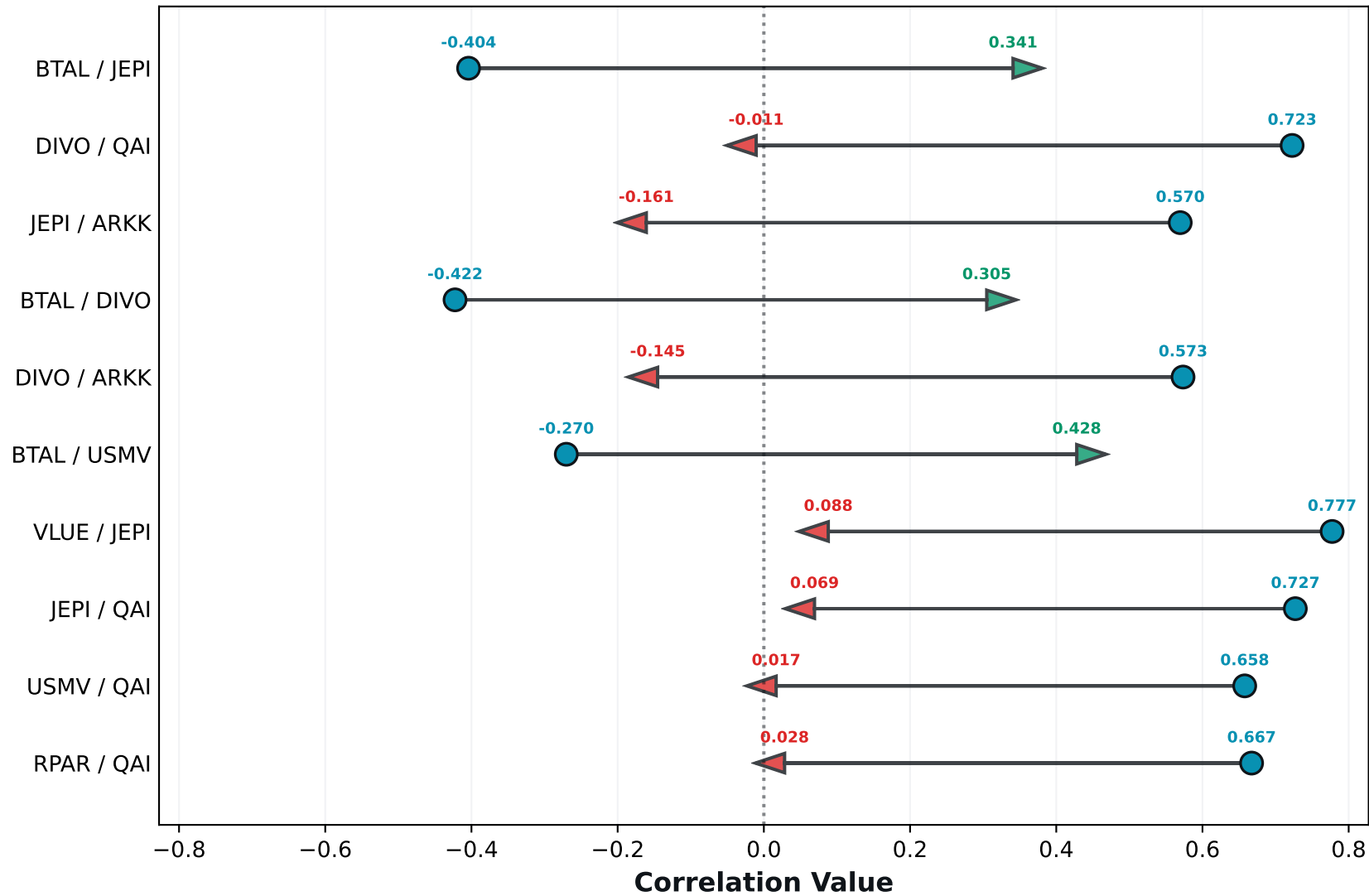
How correlation changes when factors are removed



Correlation change. Below diagonal: factor model explained the co-movement. Above diagonal: unexpected residual correlation after factor removal. Color: blue = decrease, red = increase.

Largest Correlation Changes

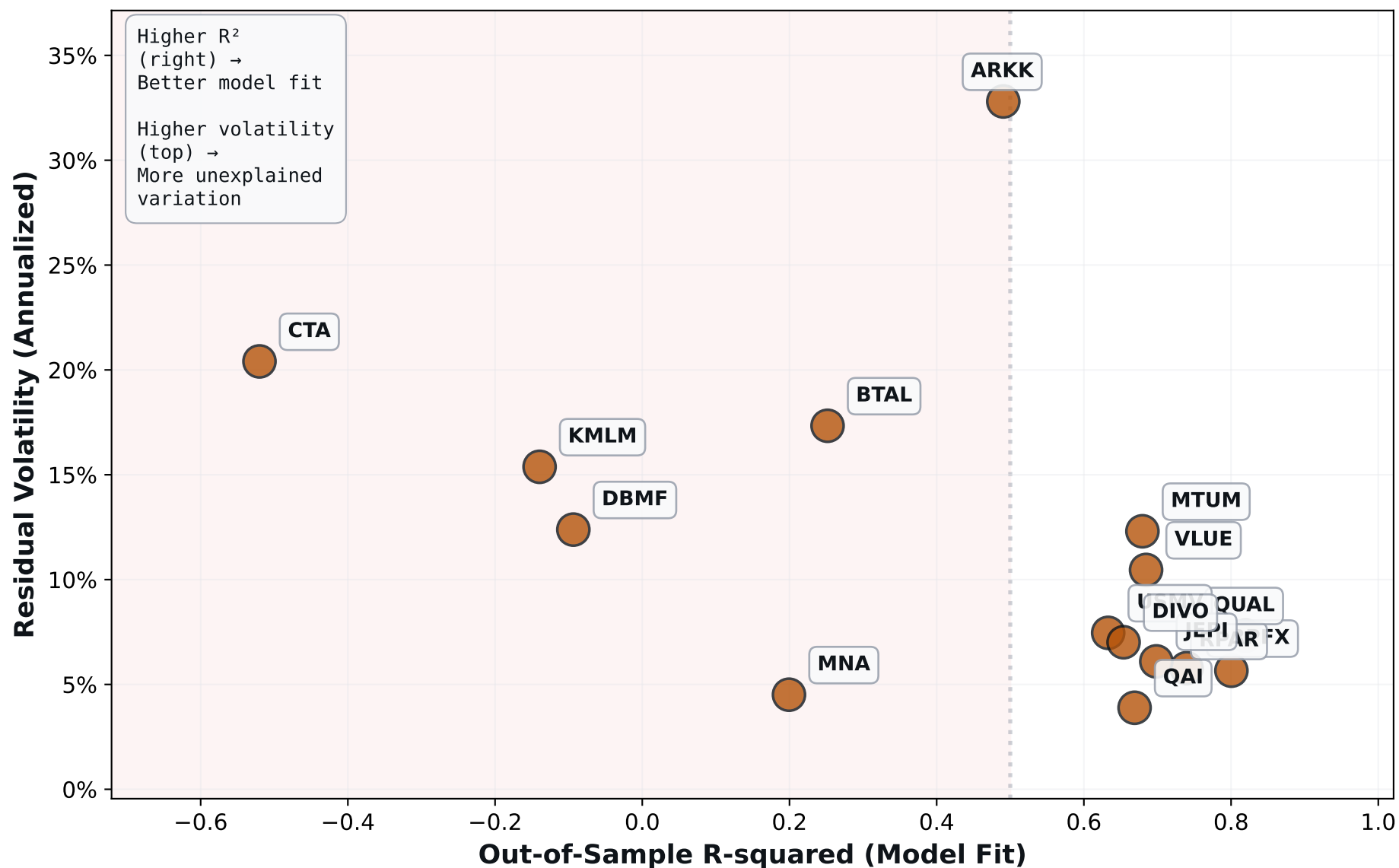
Direction and magnitude of shift from raw to residual correlation



Blue arrows: Correlation decreased after factor removal (factor-driven co-movement) • Red arrows: Correlation increased (potential missing factors)

Model Quality by Asset

Residual volatility vs. out-of-sample fit quality



Assets at bottom-left (low volatility, high fit) are well-explained by the model. Assets at top-left or bottom-middle indicate model risk.